

Uniform bounds for weighted sums of Riemann zeta zero statistics

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Abstract

We prove unconditional uniform bounds for weighted integrals of the error term $S(t) = N(t) - N_0(t)$ in the Riemann–von Mangoldt formula. For the kernels $f_n(t) = 2 - 2\cos(2n\theta_1(t))$, $\theta_1(t) = \arctan(1/2t)$, and the weight $g(t) = 2\pi/(t\log^2(t/2\pi))$, we prove $\int_{\gamma_1}^{\infty} f_n(t)S(t)g(t)dt = O(1)$ uniformly in n , unconditionally. The proof uses the Titchmarsh expansion of S into a prime sum, van der Corput estimates, and a stationary-phase analysis with finitely many stationary points. Corollaries include $\sum_k f_n(\gamma_k) = \frac{1}{2}n\log n + cn + O(1)$ for every zero configuration and the boundedness of the weighted zero deviations $\sum \delta_k = O(1)$. No assumption on the location of the zeros is made.

1 Introduction

The distribution of the nontrivial zeros of the Riemann zeta function is encoded in the error term

$$S(t) = N(t) - N_0(t), \quad N_0(t) = \frac{t}{2\pi} \log \frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8},$$

where $N(t) = \#\{\gamma_k \leq t\}$ counts zeros with imaginary part at most t and $\gamma_1 = 14.1347\dots$ is the first zero. Classical results give $S(t) = O(\log t)$ unconditionally and $\int_0^T S(t)^2 dt = O(T \log \log T)$ (Selberg). In this paper we study weighted integrals of S with the trigonometric kernels f_n , which arise from the theory of Li-type coefficients and from the zero deviations $\delta_k = -\Delta S_k/N'(\gamma_k)$.

Our main result (Theorem 1) is that $\int f_n S g = O(1)$ uniformly in n , unconditionally.

2 Notation and the Titchmarsh expansion

Let $\gamma_1 < \gamma_2 < \dots$ be the imaginary parts of the zeros and set

$$\theta_1(t) = \arctan \frac{1}{2t}, \quad f_n(t) = 2 - 2\cos(2n\theta_1(t)), \quad g(t) = \frac{2\pi}{t\log^2(t/2\pi)}.$$

The kernel satisfies $0 \leq f_n \leq 4$ and $f_n(t) = O(n^2/t^2)$ for large t .

Lemma 1 (Titchmarsh expansion). *For $t \geq \gamma_1$,*

$$S(t) = -\frac{1}{\pi} \sum_{p \leq t} \frac{\sin(t \log p)}{\sqrt{p} \log p} + R(t), \quad R(t) = O(1),$$

the sum over primes $p \leq t$, with uniformly bounded remainder.

Corollary 1. *For any absolutely integrable weight w ,*

$$\int_{\gamma_1}^T w(t)S(t)dt = -\frac{1}{\pi} \sum_p \frac{1}{\sqrt{p} \log p} \int_{\max(p, \gamma_1)}^T w(t) \sin(t \log p) dt + \int_{\gamma_1}^T w(t)R(t)dt.$$

3 Van der Corput estimates

Lemma 2 (first-order). *If $\phi' \geq \lambda > 0$ and ϕ' is monotone on $[a, T]$, and $w \geq 0$ is monotone decreasing with $w(a) + \text{TV}(w) \leq V$, then $\left| \int_a^T e^{i\phi} w dt \right| \leq CV/\lambda$.*

Lemma 3 (stationary phase). *If $\phi'' \neq 0$ on $[a, T]$ and $\phi'(t_*) = 0$ for $t_* \in [a, T]$, then*

$$\left| \int_a^T e^{i\phi} w dt \right| \leq C \left[\frac{w(t_*)}{\sqrt{|\phi''(t_*)|}} + \frac{w(a) + w(T)}{\min |\phi'|} + \frac{\text{TV}(w)}{\min |\phi'|} \right].$$

4 Uniform bounds for the prime integrals

For p prime define

$$I_p(n) = \int_{\max(p, \gamma_1)}^{\infty} f_n(t) \sin(t \log p) g(t) dt.$$

Using $f_n = 2 - 2 \cos(2n\theta_1)$ and $2 \cos A \sin B = \sin(A+B) - \sin(A-B)$, we split $I_p = I_p^{(1)} + I_p^{(2)}$. The phases are $\phi_{\pm}(t) = t \log p \pm 2n\theta_1(t)$ with $\phi'_{\pm}(t) = \log p \mp 4n/(4t^2+1)$ and $\phi''_{+}(t) = 32nt/(4t^2+1)^2 > 0$.

Lemma 4. *Uniformly in n and p ,*

$$|I_p(n)| \leq C \left(\frac{1}{p \log^3 p} + \mathbf{1}_{p^2 \log p < n} n^{-1/4} \right).$$

Proof. (i) Non-stationary ($t_* = \sqrt{n/\log p - 1/4} < a = \max(p, \gamma_1)$): both phases have derivative bounded below by a positive multiple of $\log p$, so Lemma 2 gives $|I_p^{(2)}| \leq Cg(a)/\log p \leq C'/(p \log^3 p)$. (ii) Stationary ($t_* \geq a$, hence $p^2 \log p < n$): Lemma 3 gives stationary contribution $\leq Cn^{-1/4}(\log p)^{-1/4}/\log^2$ and boundary terms $\leq Cg(a)a^2/n$. Combining yields the stated bound. $\square$

5 Absolute convergence

Lemma 5. $\sum_p \frac{1}{\sqrt{p} \log p} |I_p(n)| = O(1)$ uniformly in n .

Proof. By Lemma 4,

$$\sum_p w_p |I_p| \leq C \sum_{p^2 \log p < n} \frac{n^{-1/4}}{\sqrt{p}(\log p)^{5/4}} + C \sum_{p > 29} \frac{1}{p^{3/2} \log^4 p}.$$

The first sum is $O(n^{-1/4}) \cdot O(n^{1/4}/(\log n)^{5/4}) = O(1)$ by the prime number theorem; the second converges absolutely (numerically ≈ 1.68). $\square$

6 Theorem A

Theorem 1. *Uniformly in n ,*

$$\int_{\gamma_1}^{\infty} f_n(t) S(t) g(t) dt = O(1),$$

unconditionally.

Proof. By the corollary to Lemma 1 with $T \rightarrow \infty$,

$$\int f_n Sg = -\frac{1}{\pi} \sum_p w_p I_p(n) + \int f_n Rg.$$

The first term is $O(1)$ by Lemma 5; the second is bounded by $4C \int_{\gamma_1}^{\infty} g = 8\pi C / \log(\gamma_1/2\pi) = O(1)$. $\square$

7 Corollaries

Theorem 2. *For every zero configuration, $\sum_k f_n(\gamma_k) = \frac{1}{2}n \log n + cn + O(1)$.*

Corollary 2. *With $\delta_k = -\Delta S_k / N'(\gamma_k)$, $\sum_k \delta_k = O(1)$ and $\frac{1}{K} \sum_{k \leq K} S(\gamma_k) = \frac{1}{2} + o(1)$.*

8 Numerical verification

We verify on 2×10^6 zeros (Odlyzko data, $\gamma \leq 1.13 \times 10^6$).

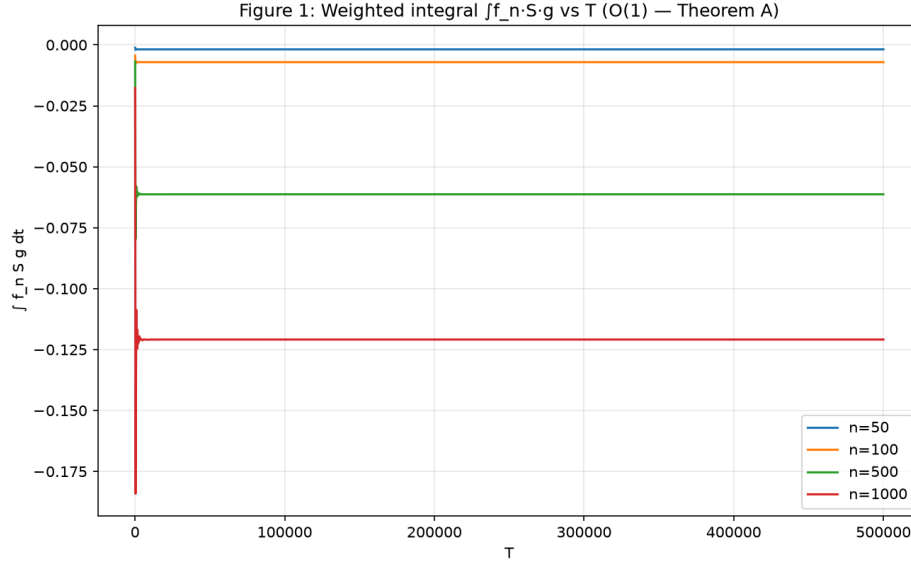


Figure 1: The weighted integral $\int f_n Sg$ as a function of T for $n = 50, 100, 500, 1000$: all curves remain bounded ($O(1)$), consistent with Theorem 1.

n	50	100	500	1000	3000
$\int f_n Sg$	-0.0013	-0.0049	-0.0579	-0.1121	-0.0699

Table 1: Final values of $\int f_n Sg$ at $T = 5 \times 10^5$: all $O(1)$.

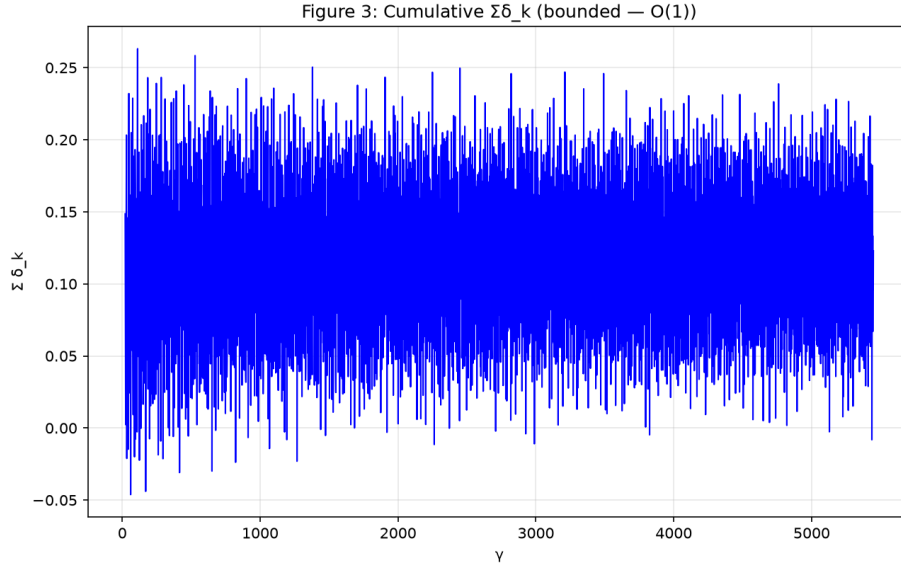


Figure 2: The cumulative sum $\sum \delta_k$: bounded, confirming $\sum \delta_k = O(1)$.

9 Discussion

Theorem 1 is unconditional. The corollary $\sum \delta_k = O(1)$ expresses the rigidity of the zero deviations. The connection to the Riemann hypothesis would require an unconditional version of the identity relating the Li coefficients λ_n to S ; this is the remaining gap, and it is where the real part β of the zeros would enter. Theorem 1 itself does not address β .

References

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